

USSR Programming Languages Timeline

1955

Address Programming Language



Address Programming Language

Early Soviet high-level language by Kateryna Yushchenko. Used for economic, military, aviation, and early space calculations.

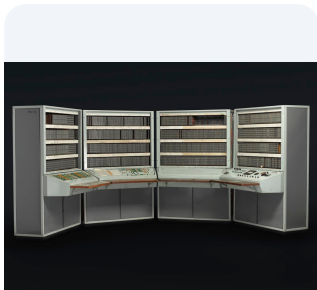
Key points

- Created in Kyiv
- Advanced address/pointer ideas
- Applied to early technical planning

Refs [1][2]

1958–63

ALMO



ALMO

Machine-oriented algorithmic system used in compiler and systems work around major Soviet installations.

Key points

- Compiler-oriented design
- Close to machine structure
- Linked to BESM-era systems work

Refs [2][3]

1962–63

LYaPAS



LYaPAS

Tomsk language for non-numeric and logic-style processing, later adapted to the Soviet GOST 10859 character set.

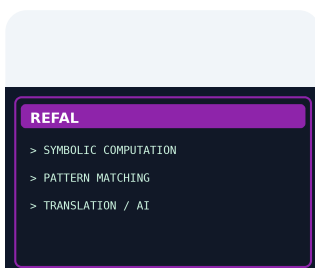
Key points

- Developed in Tomsk
- Non-numeric processing focus
- Adapted to GOST 10859

Refs [4]

1966–68

Refal



Refal

Symbolic pattern-matching language used for translation, formula manipulation, and early AI-style work.

Key points

- Created by Valentin Turchin
- Strong symbolic manipulation
- Useful for translation and AI

Refs [5][6]

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1978–79

Rapira



Rapira

Russian-syntax educational language for Agat and Soviet PDP-11-compatible systems.

Key points

- Russian-style keywords
- Popular in education
- Ran on Agat and DVK-class machines

Refs [7]

1980s

Robik



Robik

Child-focused descendant of Rapira designed for primary-school teaching.

Key points

- Made for younger learners
- Simplified educational focus
- School-oriented rather than industrial

Refs [7]

1986

Buran toolchain



Buran toolchain

PROL2, DIPOL, and LAKS supported onboard, ground, and modelling software during the Buran shuttle effort.

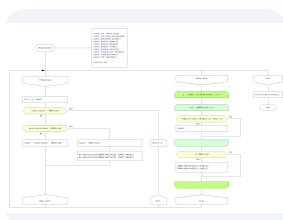
Key points

- Multiple specialist languages
- Aerospace mission software
- Set the stage for DRAGON

Refs [8][9]

1996

DRAGON



DRAGON

Visual language completed after Buran and later reported in aerospace, oil, and nuclear-power procedures.

Key points

- Diagram-based notation
- Built for clarity and safety
- Later used beyond space projects

Refs [9][10]

References

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